



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science and Technology

Department of Computer Science

CSC 3215: Web Technologies

Semester: Summer 2025-26

Group No: 04

Section:

U

Group Members:

SL.	Student ID	Name	Role	Signature
1	23-53712-3	Shantunu Barua	Group Leader	Shantunu
2	23-55565-3	Md. Mahamudul Parvej	Member	Parvej
3	23-53391-3	Israt Jahan Suporna	Member	Suporna
4	23-55678-8	Tanaka Rahman	Member	Tanaka

Project Title

Rescue Management System

Short Description

Rescue Management System is a centralized role-based web application designed to coordinate emergency assistance among Admin, Volunteer, Witness, and Help Seeker users. The system manages rescue requests, volunteer assignment and activity, incident reporting with evidence, donations, resource requests, feedback, rescue reports, and role-based notifications through an MVC-style PHP and MySQL application.

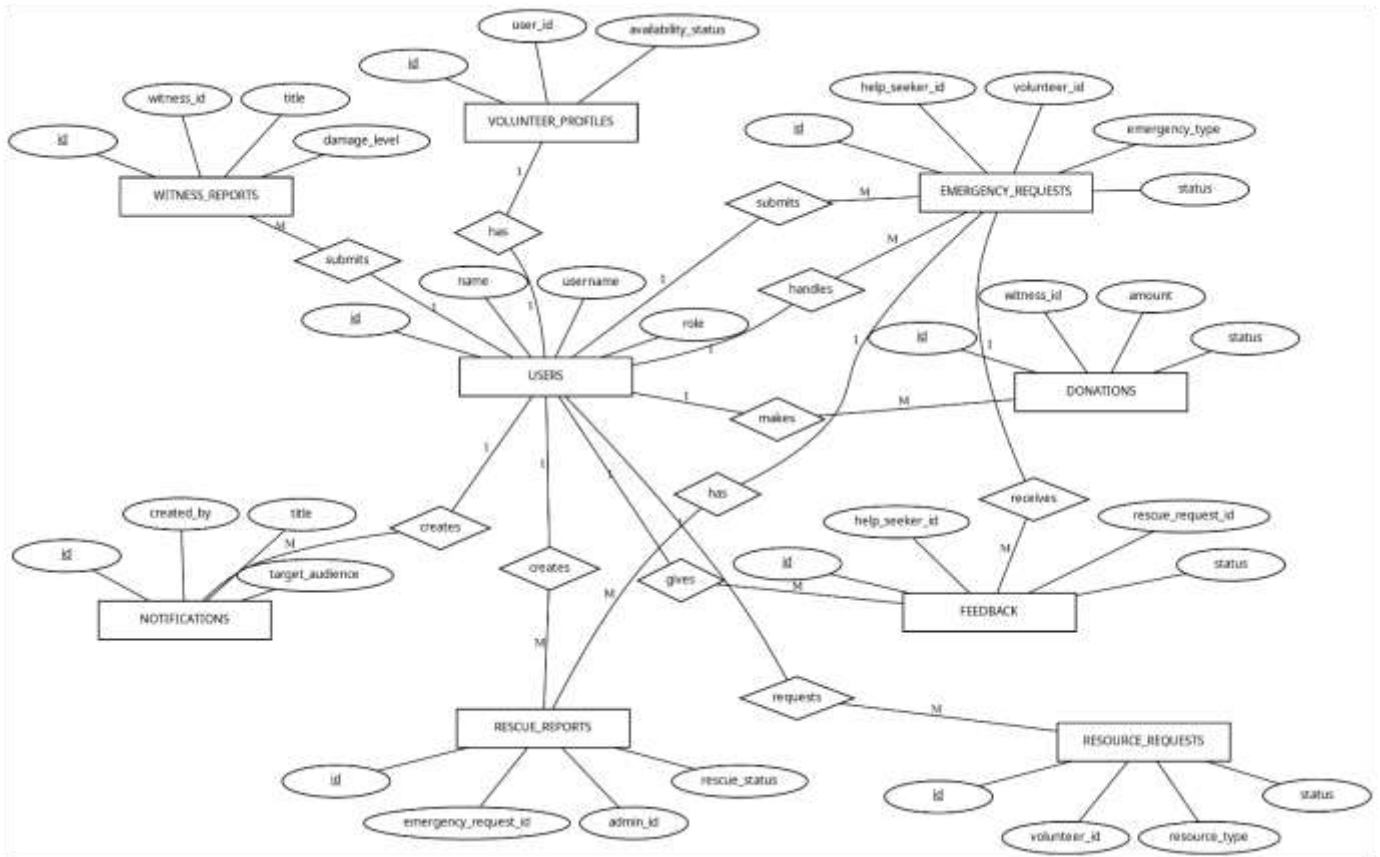
Introduction

Emergency situations require quick communication, organized response, and clear information sharing. A disconnected process can make it difficult for people seeking help, volunteers, witnesses, and administrators to coordinate rescue activities. This project provides one web platform where a Help Seeker can submit an emergency request, a Volunteer can accept and manage rescue activity, a Witness can report incidents and provide supporting evidence, and an Admin can supervise the overall workflow.

Background Study

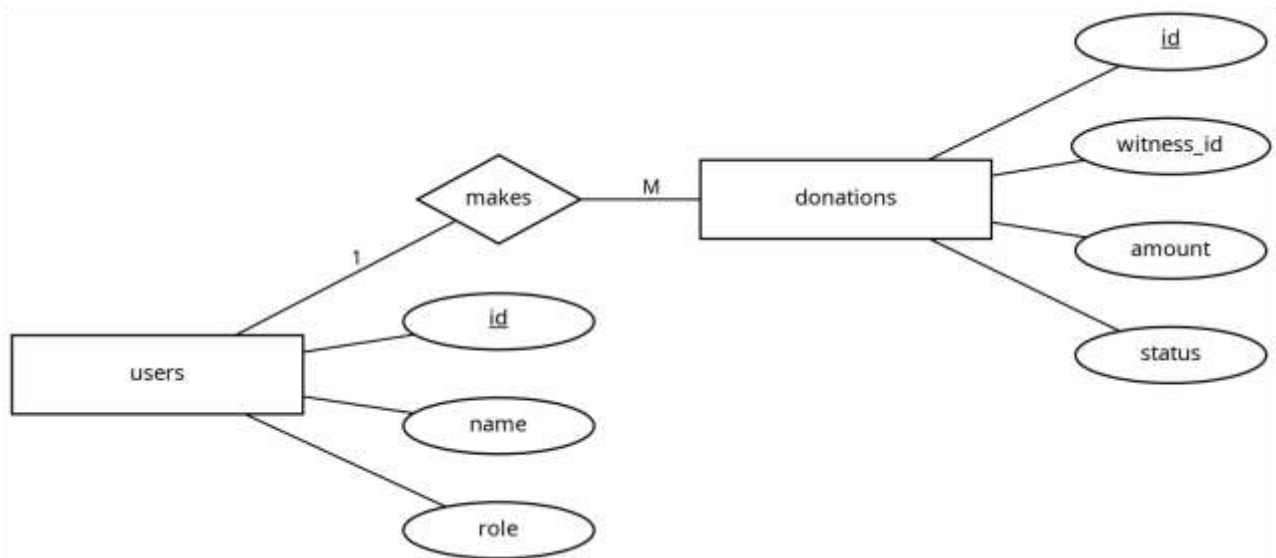
The project is implemented with PHP, MySQL/MariaDB, HTML, CSS, and JavaScript. It follows an MVC-style structure with separate models, controllers, and views, while index.php works as the front controller and router. The application also uses session-based authentication, role guards, server-side validation, CSRF protection, AJAX/JSON search, responsive interface design, and controlled evidence-file uploads.

ER Diagram:



Normalization:

1. users $\rightarrow$ donations



Relation: One to many

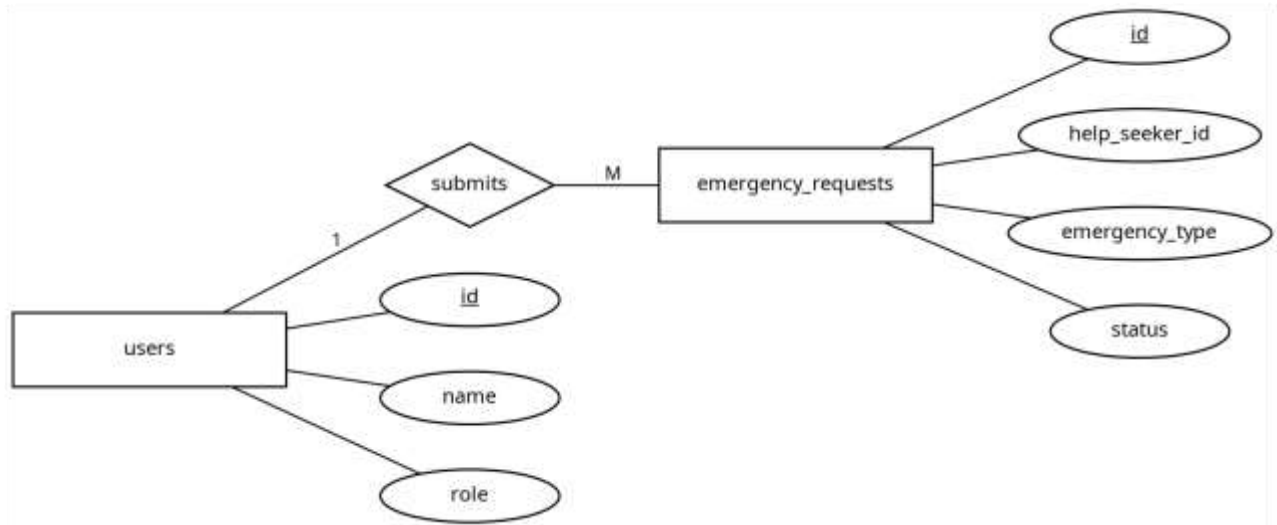
UNF: id, name, username, email, phone, password, role, created_at, updated_at, id, witness_id, amount, donation_type, payment_method, transaction_id, message, status, created_at

1NF: No multivalued attributes

users: id, name, username, email, phone, password, role, created_at, updated_at

donations: id, witness_id, amount, donation_type, payment_method, transaction_id, message, status, created_at

2. users → emergency_requests



Relation: One to many

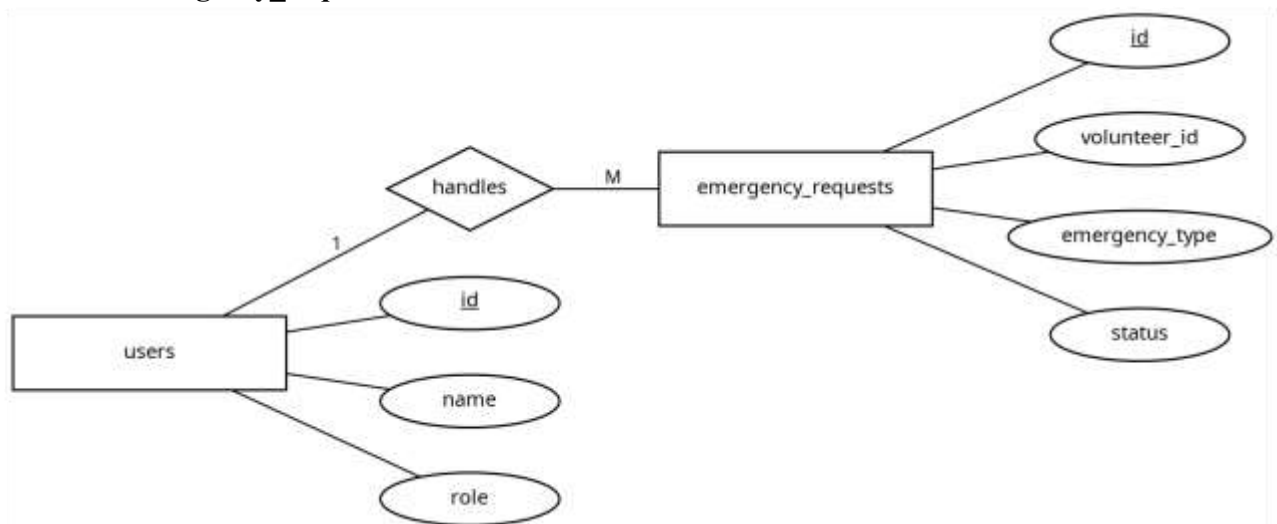
UNF: id, name, username, email, phone, password, role, created_at, updated_at, id, help_seeker_id, emergency_type, location, description, priority, victim_type, victim_information, victim_count, contact_information, status, created_at, updated_at, volunteer_id, accepted_at

1NF: No multivalued attributes

users: id, name, username, email, phone, password, role, created_at, updated_at

emergency_requests: id, help_seeker_id, emergency_type, location, description, priority, victim_type, victim_information, victim_count, contact_information, status, created_at, updated_at, volunteer_id, accepted_at

3. users → emergency_requests



Relation: One to many

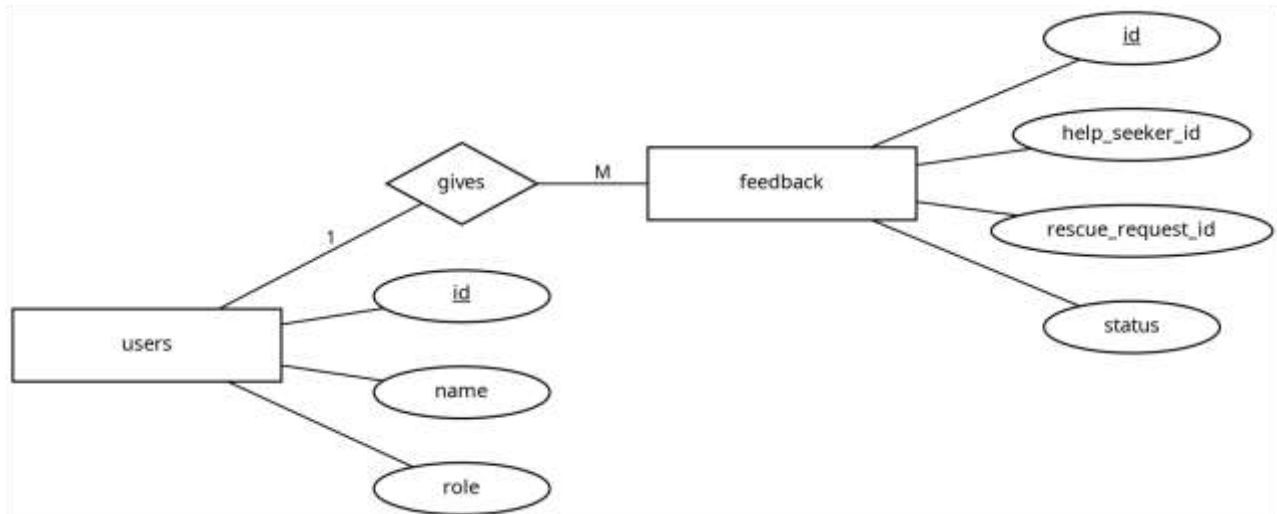
UNF: id, name, username, email, phone, password, role, created_at, updated_at, id, help_seeker_id, emergency_type, location, description, priority, victim_type, victim_information, victim_count, contact_information, status, created_at, updated_at, volunteer_id, accepted_at

1NF: No multivalued attributes

users: id, name, username, email, phone, password, role, created_at, updated_at

emergency_requests: id, help_seeker_id, emergency_type, location, description, priority, victim_type, victim_information, victim_count, contact_information, status, created_at, updated_at, volunteer_id, accepted_at

4. users → feedback



Relation: One to many

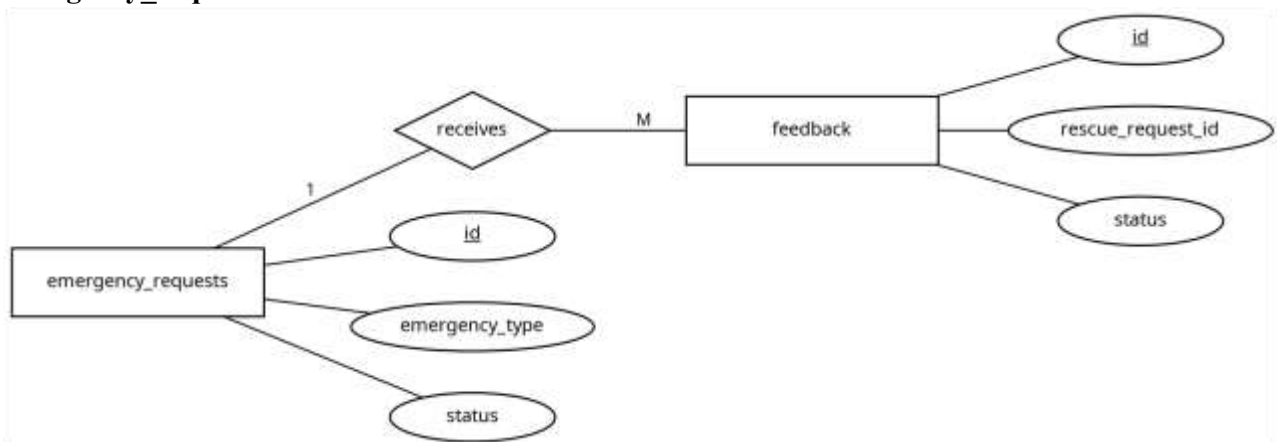
UNF: id, name, username, email, phone, password, role, created_at, updated_at, id, help_seeker_id, rescue_request_id, message, status, created_at, updated_at

1NF: No multivalued attributes

users: id, name, username, email, phone, password, role, created_at, updated_at

feedback: id, help_seeker_id, rescue_request_id, message, status, created_at, updated_at

5. emergency_requests → feedback



Relation: One to many

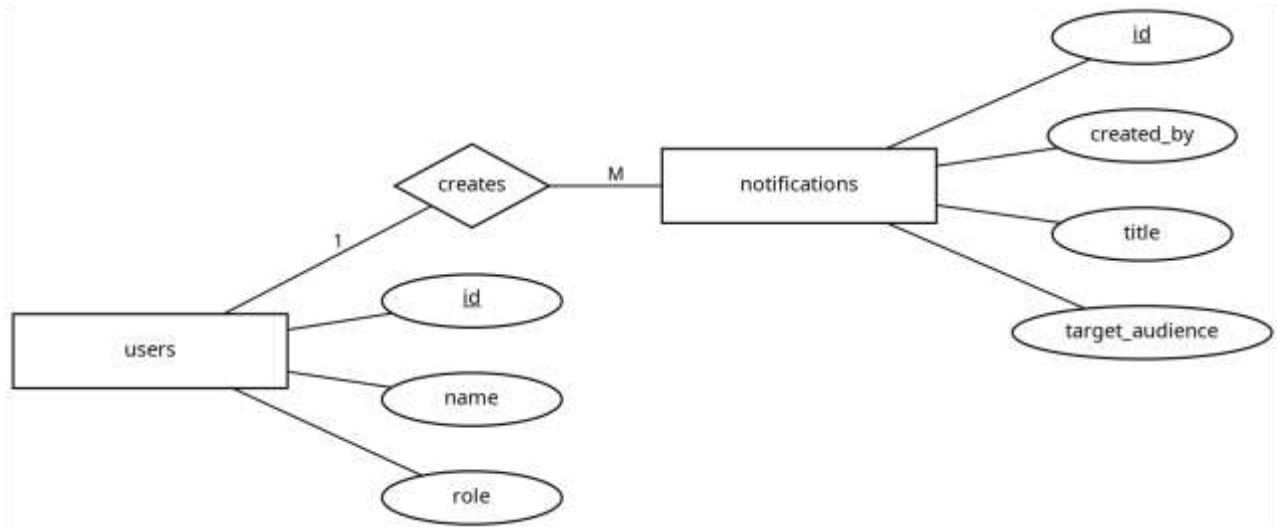
UNF: id, help_seeker_id, emergency_type, location, description, priority, victim_type, victim_information, victim_count, contact_information, status, created_at, updated_at, volunteer_id, accepted_at, id, help_seeker_id, rescue_request_id, message, status, created_at, updated_at

1NF: No multivalued attributes

emergency_requests: id, help_seeker_id, emergency_type, location, description, priority, victim_type, victim_information, victim_count, contact_information, status, created_at, updated_at, volunteer_id, accepted_at

feedback: id, help_seeker_id, rescue_request_id, message, status, created_at, updated_at

6. users → notifications



Relation: One to many

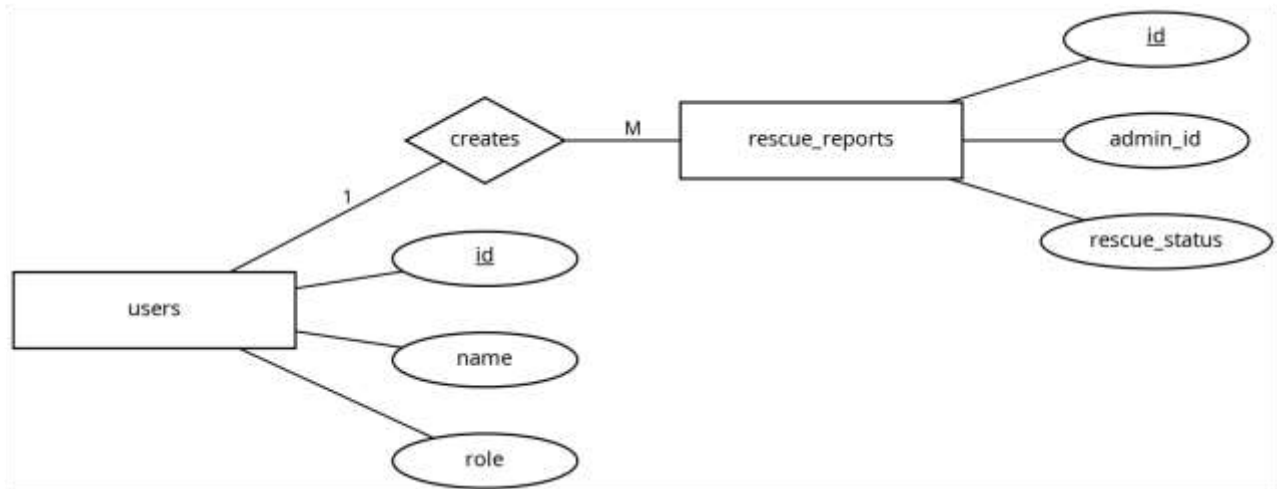
UNF: id, name, username, email, phone, password, role, created_at, updated_at, id, created_by, title, message, alert_type, target_audience, status, created_at, updated_at

1NF: No multivalued attributes

users: id, name, username, email, phone, password, role, created_at, updated_at

notifications: id, created_by, title, message, alert_type, target_audience, status, created_at, updated_at

7. users → rescue_reports



Relation: One to many

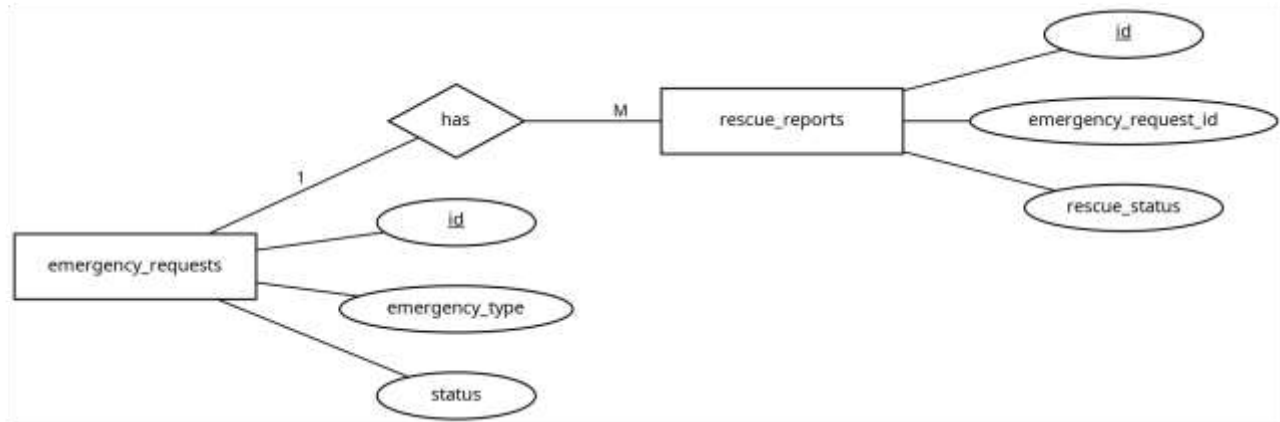
UNF: id, name, username, email, phone, password, role, created_at, updated_at, id, emergency_request_id, admin_id, rescue_status, description, created_at, updated_at

1NF: No multivalued attributes

users: id, name, username, email, phone, password, role, created_at, updated_at

rescue_reports: id, emergency_request_id, admin_id, rescue_status, description, created_at, updated_at

8. emergency_requests → rescue_reports



Relation: One to many

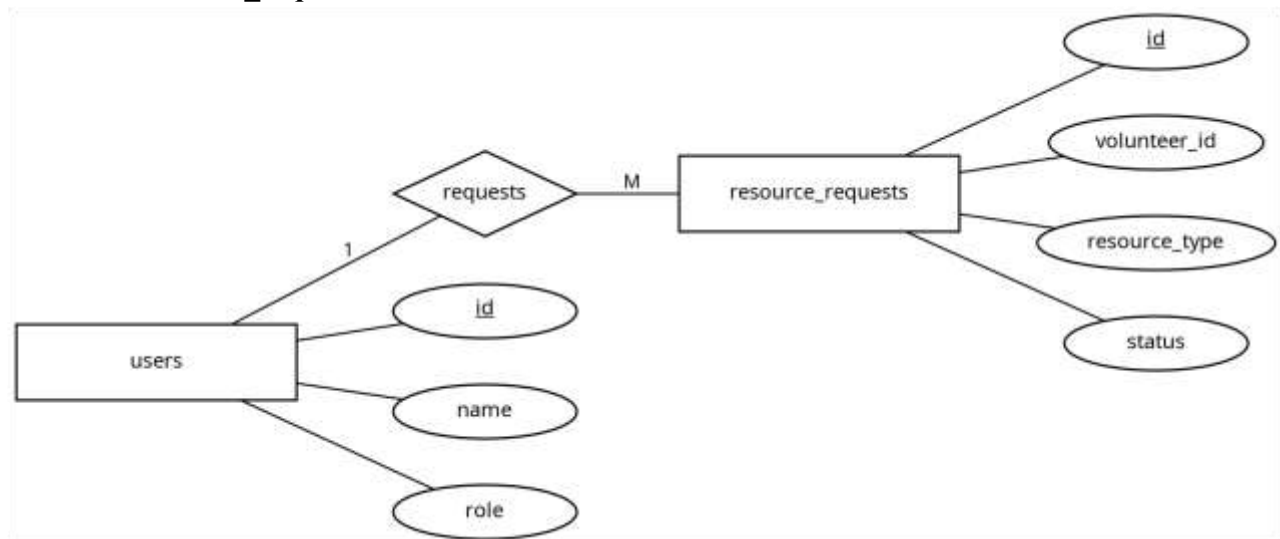
UNF: id, help_seeker_id, emergency_type, location, description, priority, victim_type, victim_information, victim_count, contact_information, status, created_at, updated_at, volunteer_id, accepted_at, id, emergency_request_id, admin_id, rescue_status, description, created_at, updated_at

1NF: No multivalued attributes

emergency_requests: id, help_seeker_id, emergency_type, location, description, priority, victim_type, victim_information, victim_count, contact_information, status, created_at, updated_at, volunteer_id, accepted_at

rescue_reports: id, emergency_request_id, admin_id, rescue_status, description, created_at, updated_at

9. users → resource_requests



Relation: One to many

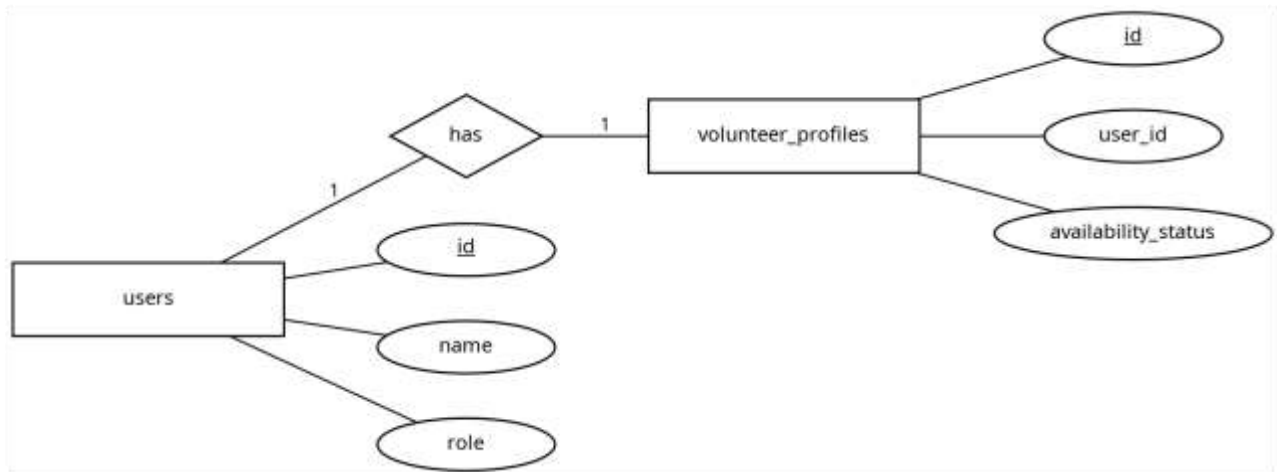
UNF: id, name, username, email, phone, password, role, created_at, updated_at, id, volunteer_id, resource_type, quantity, description, status, created_at, updated_at

1NF: No multivalued attributes

users: id, name, username, email, phone, password, role, created_at, updated_at

resource_requests: id, volunteer_id, resource_type, quantity, description, status, created_at, updated_at

10. users → volunteer_profiles



Relation: One to one

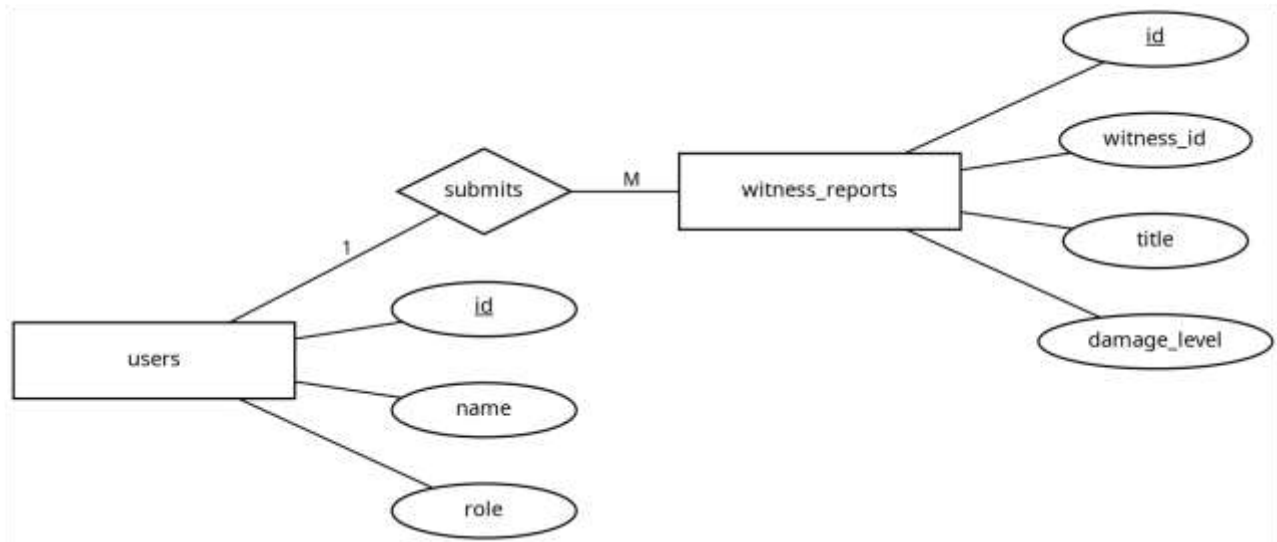
UNF: id, name, username, email, phone, password, role, created_at, updated_at, id, user_id, address, blood_group, experience, skills, emergency_contact, availability_status, created_at, updated_at

1NF: No multivalued attributes

users: id, name, username, email, phone, password, role, created_at, updated_at

volunteer_profiles: id, user_id, address, blood_group, experience, skills, emergency_contact, availability_status, created_at, updated_at

11. users → witness_reports



Relation: One to many

UNF: id, name, username, email, phone, password, role, created_at, updated_at, id, witness_id, title, description, damage_level, incident_type, location, incident_date, evidence_file, status, created_at, updated_at

1NF: No multivalued attributes

users: id, name, username, email, phone, password, role, created_at, updated_at

witness_reports: id, witness_id, title, description, damage_level, incident_type, location, incident_date, evidence_file, status, created_at, updated_at

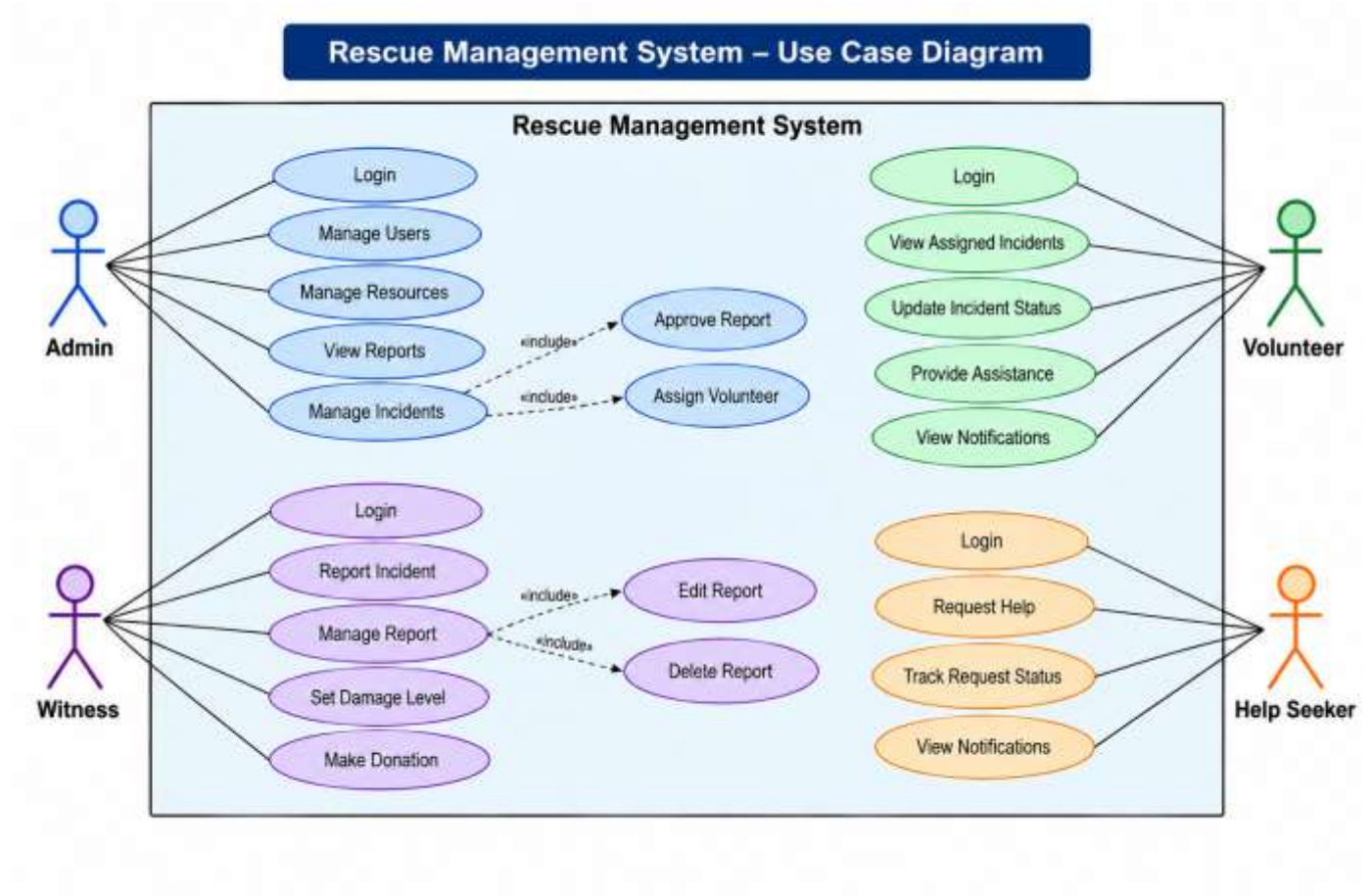
Total Tables:

1. **users:** id, name, username, email, phone, password, role, created_at, updated_at
2. **donations:** id, witness_id, amount, donation_type, payment_method, transaction_id, message, status, created_at
3. **users:** id, name, username, email, phone, password, role, created_at, updated_at
4. **emergency_requests:** id, help_seeker_id, emergency_type, location, description, priority, victim_type, victim_information, victim_count, contact_information, status, created_at, updated_at, volunteer_id, accepted_at
5. **users:** id, name, username, email, phone, password, role, created_at, updated_at
6. **emergency_requests:** id, help_seeker_id, emergency_type, location, description, priority, victim_type, victim_information, victim_count, contact_information, status, created_at, updated_at, volunteer_id, accepted_at
7. **users:** id, name, username, email, phone, password, role, created_at, updated_at
8. **feedback:** id, help_seeker_id, rescue_request_id, message, status, created_at, updated_at
9. **emergency_requests:** id, help_seeker_id, emergency_type, location, description, priority, victim_type, victim_information, victim_count, contact_information, status, created_at, updated_at, volunteer_id, accepted_at
10. **feedback:** id, help_seeker_id, rescue_request_id, message, status, created_at, updated_at
11. **users:** id, name, username, email, phone, password, role, created_at, updated_at
12. **notifications:** id, created_by, title, message, alert_type, target_audience, status, created_at, updated_at
13. **users:** id, name, username, email, phone, password, role, created_at, updated_at
14. **rescue_reports:** id, emergency_request_id, admin_id, rescue_status, description, created_at, updated_at
15. **emergency_requests:** id, help_seeker_id, emergency_type, location, description, priority, victim_type, victim_information, victim_count, contact_information, status, created_at, updated_at, volunteer_id, accepted_at
16. **rescue_reports:** id, emergency_request_id, admin_id, rescue_status, description, created_at, updated_at
17. **users:** id, name, username, email, phone, password, role, created_at, updated_at
18. **resource_requests:** id, volunteer_id, resource_type, quantity, description, status, created_at, updated_at
19. **users:** id, name, username, email, phone, password, role, created_at, updated_at
20. **volunteer_profiles:** id, user_id, address, blood_group, experience, skills, emergency_contact, availability_status, created_at, updated_at
21. **users:** id, name, username, email, phone, password, role, created_at, updated_at
22. **witness_reports:** id, witness_id, title, description, damage_level, incident_type, location, incident_date, evidence_file, status, created_at, updated_at.

Finalizations:

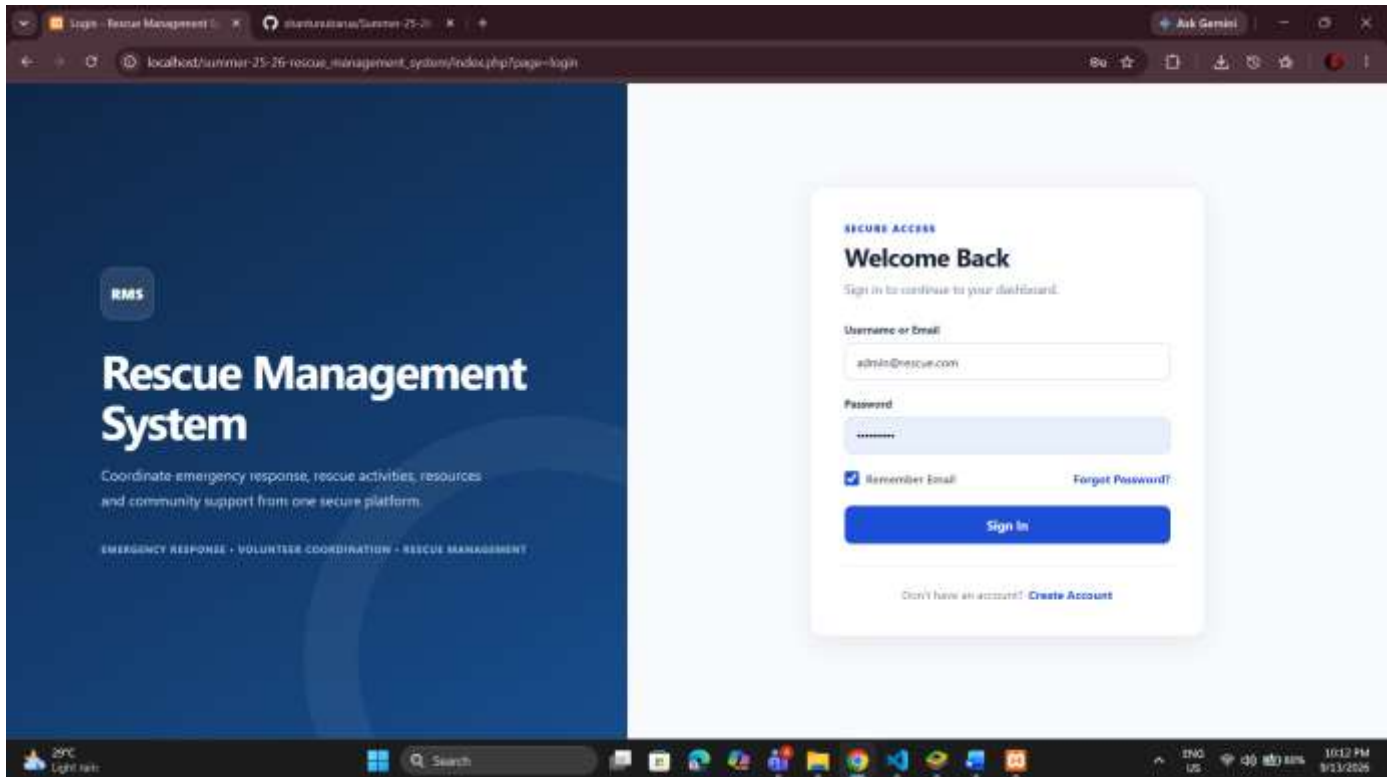
1. users: id, name, username, email, phone, password, role, created_at, updated_at
2. donations: id, witness_id, amount, donation_type, payment_method, transaction_id, message, status, created_at
3. emergency_requests: id, help_seeker_id, emergency_type, location, description, priority, victim_type, victim_information, victim_count, contact_information, status, created_at, updated_at, volunteer_id, accepted_at
4. feedback: id, help_seeker_id, rescue_request_id, message, status, created_at, updated_at
5. notifications: id, created_by, title, message, alert_type, target_audience, status, created_at, updated_at
6. rescue_reports: id, emergency_request_id, admin_id, rescue_status, description, created_at, updated_at
7. resource_requests: id, volunteer_id, resource_type, quantity, description, status, created_at, updated_at
8. volunteer_profiles: id, user_id, address, blood_group, experience, skills, emergency_contact, availability_status, created_at, updated_at
9. witness_reports: id, witness_id, title, description, damage_level, incident_type, location, incident_date, evidence_file, status, created_at, updated_at

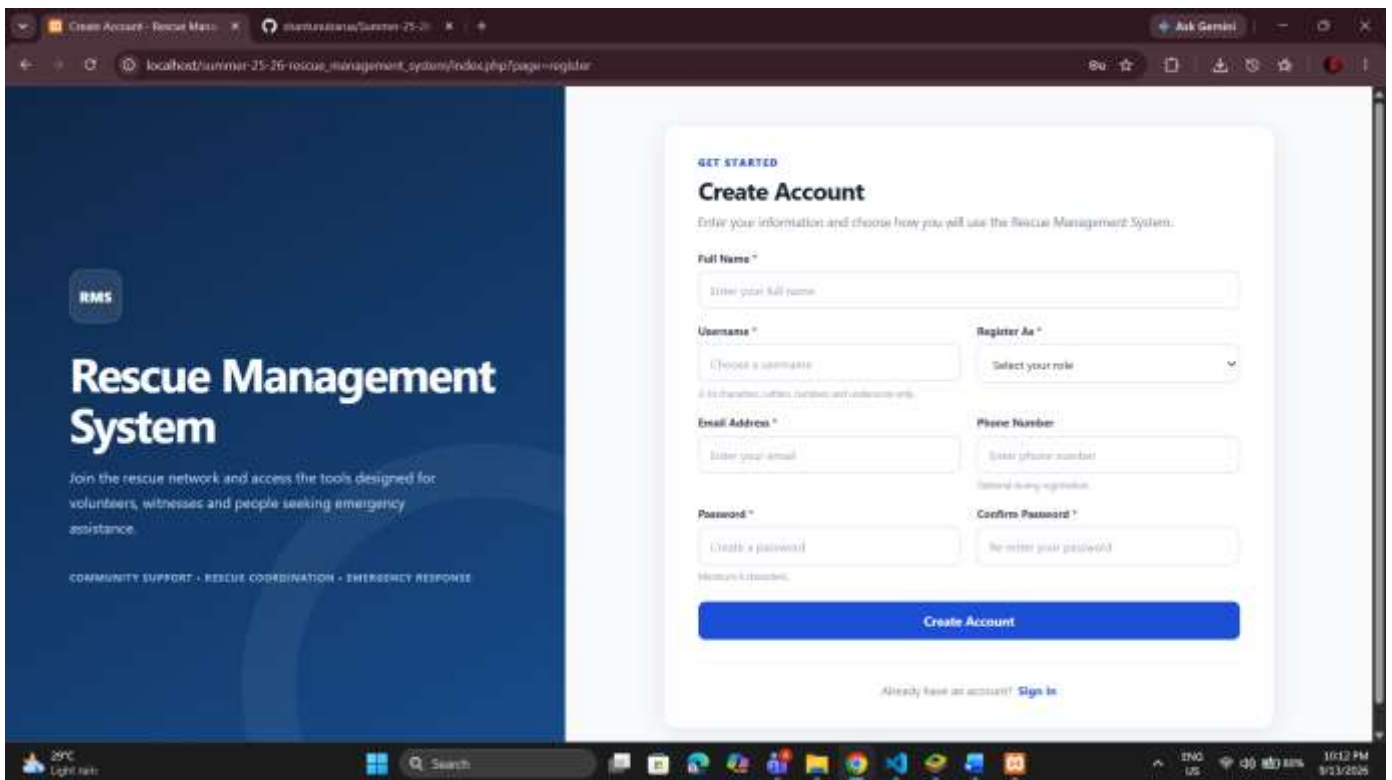
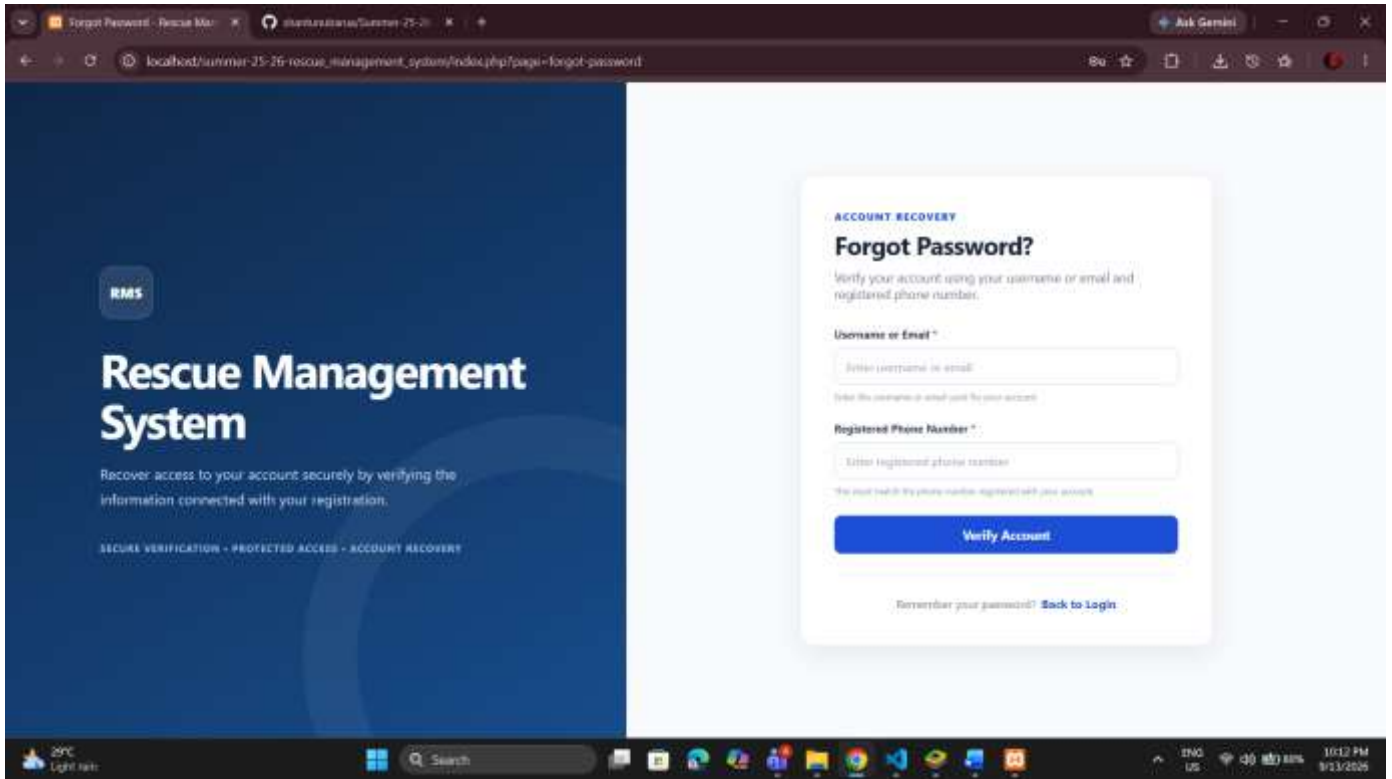
Use Case Diagram:



UI (Figma/Sketch)

The application uses separate role-based dashboards and a shared responsive stylesheet. The following representative overview is based on the implemented view modules and shows the major actions exposed to each ro





RMS

Rescue Management System

SIGNED IN AS
System Admin

ADMIN PANEL

Dashboard

Notifications

Feedback

Witness Reports

Rescue Reports

Resource Requests

Donations

Change Password

Logout

ADMINISTRATION OVERVIEW

Admin Dashboard

Welcome back, System Admin. Review system activity and manage the Rescue Management System from one place.

USER OVERVIEW

Registered Users

Current registered user accounts by role:

ROLE

WITNESS

WIT

HELP

HELP SEEKER

1

2

1

Registered witness accounts available in the system.

Registered witness accounts in the system.

Registered help seeker accounts using rescue services.

SYSTEM ACTIVITY

Management Overview

Five months currently managed by the administration

WIT

WIT

WIT

WIT

4

2

1

2

Notifications

Feedback

Rescue Reports

Emergency Requests

Manage system announcements and important user alerts.

Review feedback submitted after rescue activities.

Review and manage official rescue operations reports.

Track emergency rescue requests recorded in the system.

View Notifications

Review Feedback

View Reports

View Requests

29°C

Light rain

Search

10:12 PM

8/13/2025

RMS

Rescue Management System

SIGNED IN AS
Parvej

HELP SEEKER PANEL

Dashboard

Request Rescue

My Requests

Nearby Volunteers

My Profile

Change Password

Logout

HELP SEEKER OVERVIEW

Help Seeker Dashboard

Welcome back, Parvej. Request emergency assistance, track your rescue requests and find available volunteers.

My Profile

Request Rescue

NEED RESCUE ASSISTANCE?

SO\$

Create an Emergency Request

Request Rescue

Submit your location, emergency type, priority, victim information and contact details so volunteers can respond.

QUICK ACCESS

Help Seeker Actions

EMERGENCY

TRACKING

VOLUNTEERS

ACCOUNT

Request Rescue

My Requests

Nearby Volunteers

My Profile

Submit a new emergency rescue request with victim and priority information.

Search and track your submitted rescue requests and their current status.

Find available volunteers using the project's local area-based matching.

Review and update your personal contact and account information.

Create Request...

View Requests...

Find Volunteers...

View Profile...

ADMIN ALERTS

Notifications & Alerts

29°C

Light rain

Search

10:13 PM

8/13/2025

RMS

Rescue Management System

SIGNED IN AS
Tanaka Rahman

WITNESS PANEL

Dashboard

Report Incident

My Reports

Make Donation

My Donations

My Profile

Change Password

Logout

WITNESS OVERVIEW

My ProfileReport IncidentMake Donation

Witness Dashboard

Welcome back, Tanaka Rahman. Track incidents, donations and admin alerts from one place.

INCIDENT REPORTS

4

Enter submitted reports

View Reports

CRITICAL REPORTS

0

Reports marked critical

DONATIONS

2

Completed donations

View History

TOTAL DONATED

130,000.00

Total contribution amount

ADMIN ALERTS

Notifications

Flood emergency alert

Emergency

Rescue operation active in zone A

Address: 40 - 25 Aug 2024 10:11 PM

Witness Test

Important

This is for witnesses.

Address: Witness - 07 Sep 2024 10:48 PM

test

Normal

test message

Address: 40 - 11 Sep 2024 11:45 AM

29°C

Light rain

Search

WhatsApp

10:12 PM

9/13/2024

RMS

Rescue Management System

SIGNED IN AS
Suparna

VOLUNTEER PANEL

Dashboard

Emergency Requests

My Rescue Activities

My Availability

My Profile

Resource Request

My Resource Requests

Change Password

Logout

VOLUNTEER OPERATIONS

My Profile

Volunteer Dashboard

Welcome back, Suparna. Manage rescue activities, availability and resource requests.

CURRENT AVAILABILITY

Rescue Availability

AVAILABLE

Your availability determines whether you can respond to new rescue requests.

Update Availability

TOTAL RESCUE ACTIVITIES

1

All emergency requests accepted by your volunteer account.

View Activities

RESCUE PROGRESS

My Activity Overview

ADD

0

Rescue requests assigned and waiting to be started.

REQUESTS

DOING

0

Rescue operations currently in progress.

OPERATIONS

DONE

1

Rescue operations successfully completed.

COMPLETED

EXPLORE ACTIONS

Volunteer Actions

REQUEST

Emergency Requests

ACTIVITY

My Rescue Activities

AVAILABILITY

Availability Status

29°C

Partly cloudy

Search

WhatsApp

10:55 PM

9/13/2024

Types of Users: 1.Admin (Shantunu Barua, 23-53712-3)
2.Volunteer(Israt Jahan Suporna, 23-53391-3)
3.Witness (Tanaka Rahman, 23-55678-3)
4.Help Seeker (Md. Mahamudul Parvej, 23-55565-3)

Common Features (Available to All Users)

- **Authentication**
 1. Login to the system
 2. Logout from the system
 3. User registration
- **Account Management**
 1. Change or reset password
 2. Manage profile information (view, edit, delete)
- **Dashboard**
 1. Access a personalized dashboard after login

Specific Features by User Type

- **Features of Admin**
 1. Notification and Alert
 2. Feedback
 3. Rescue Reports

- **Features of Volunteer**
 1. Profile
 2. Availability Status
 3. Resource Request
- **Features of Witness**
 1. Incident Reporting
 2. Donate Items
 3. Damage Level
- **Features of Help Seeker**
 1. Emergency Request
 2. Volunteer Availability
 3. Priority Assistance